

Empirical Proof Record: VALIDATED

2026-05-23T23:09:43.568246+00:00

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Proof ID: 0414000a43d5c91ce590b02eb0d986f9...

Created: 2026-05-23T23:09:43.568246+00:00

1. Claim

Across 30 synthetic (x, y) sample pairs of size 50 per group drawn from a rotation of normal, log-normal, and Cauchy distributions with varying shift parameters (seed=20260518), “scipy.stats.mannwhitneyu(use_continuity=True)” and “pingouin.mwu” produce Mann-Whitney U statistics AND two-sided p-values that agree to $\leq 1e-09$ across every pair. (RFC 0002 Phase E1: cross-framework validation; first non-parametric variant.)

- **Operationalisation:** Generate N (x, y) pairs cycling through three distribution shapes (Normal(0,1), LogNormal(0,1), Cauchy(0,1)) with a sweep of location shifts. For each pair, compute Mann-Whitney U + two-sided p under both backends with use_continuity=True. Compute max_abs_diff = max over (pair, statistic $\in \{U, p\}$) of |stat_scipy - stat_pingouin|. VALIDATED iff max_abs_diff $\leq$ tolerance.
- **Threshold:** max_absolute_mannwhitneyu_difference $\leq 1e-09$ U_or_p
- **H0:** scipy and pingouin disagree on Mann-Whitney U OR the two-sided p-value by more than the floating-point tolerance under matched continuity settings.
- **H1:** Both backends compute identical U AND p to floating-point tolerance across every pair.

2. Verdict

VALIDATED — observed 0.0

30 pairs $\times$ 2 backends $\times$ 2 statistics; max pairwise $|\Delta| = 0.000e+00$ on none ($\leq 1e-09$ tol)

3. Evidence

Pillar	Statistic	Value	95% CI	I
mann_whitney_u_scipy_vs_pinguin	max_absolute_mannwhitneyu_difference	0.0	—	s

4. Pre-registration

- Registered at: 2026-05-23T23:09:43.362343+00:00
- Config hash: ad3e15120cc852c18795179166db8f78...
- Data hash: ad3e15120cc852c18795179166db8f78...

5. Signature

37e2c5b06b64aa7cbd534da2087ca37698dd723109f3094c598efdb2cb10288e